Software Requirements Specification (SRS)

Human Milk Bank Management System (HMBMS)

Document Control

Item	Details
Project Title	Human Milk Bank Management System
Client	Makati Human Milk Bank
Prepared By	
Course	*CSS152P Software Engineering 2*
Date	
Version	1.0

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1. Introduction

1.1 Purpose

This document provides a detailed description of the software requirements for the Human Milk Bank Management System (HMBMS), which is intended to be developed for the Makati Human Milk Bank. It serves as a reference document for developers, testers, and project stakeholders.

1.2 Scope

The HMBMS is a web-based system designed to digitize and streamline core milk bank operations, including:

- Donor and beneficiary registration and management
- Milk collection, lab testing, pasteurization tracking, and dispensing
- Real-time inventory management of pasteurized milk batches
- Automated report generation (daily, weekly, monthly, and annual)
- SMS notifications to registered beneficiaries when milk becomes available

The system replaces the existing manual logbook process, which is prone to delays, data entry errors, and difficult record retrieval.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
HMBMS	Human Milk Bank Management System
SMS	Short Message Service (used for beneficiary notifications)
NICU	Neonatal Intensive Care Unit
LGU	Local Government Unit
Donor	A lactating mother who donates expressed breast milk
Beneficiary	An infant or mother who receives pasteurized donor milk
Supsup Todo	Community/mobile-based milk collection program
Mom's Act	Household-based milk collection program
Milky Way	Hospital-based milk collection program
Batch ID	Unique identifier assigned to a pooled milk batch
Pasteurization	Heat treatment process to destroy pathogens in donor milk
Cold Chain	Temperature-controlled supply chain for milk transport
Admin/Staff	Milk bank personnel authorized to access the system
CRUD	Create, Read, Update, Delete (standard data operations)

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2. Overall Description

2.1 Product Perspective

HMBMS is a standalone web-based system replacing the milk bank's manual logbook process. It uses a centralized database accessible through authenticated web browsers on both desktop computers and mobile devices. An SMS module is integrated to notify beneficiaries of milk availability. Data synchronization functionality is included to enable access by staff deployed in partner hospitals.

2.2 Product Functions

- User authentication and role management
- Donor registration and profile management
- Milk collection recording (all three programs: Supsup Todo, Milky Way, Mom's Act)
- Pre-pasteurization and post-pasteurization lab result recording and batch status tracking
- Inventory management of pasteurized milk batches
- Beneficiary registration and dispensing transaction recording
- Automated SMS notifications to beneficiaries
- Automated report generation (collection, pasteurization, dispensing)
- Data synchronization between milk bank and hospital workstations

2.3 User Classes and Characteristics

User Class	Description
System Admin/Staff	Milk bank nurses and midwives managing all records, inventory, donors, beneficiaries, and reports.
Hospital Staff	Milk bank staff deployed at partnered hospitals who record Milky Way collections remotely.
Beneficiaries (SMS only)	Mothers/guardians of recipient infants who receive SMS notifications. No system login required.

2.4 Operating Environment

- Web Browser (Chrome, Edge, Firefox)
- Server-side: PHP / Python / Java (hypothetical)
- Database: MySQL / PostgreSQL

2.5 Design and Implementation Constraints

- The system must comply with the **Data Privacy Act of 2012 (RA 10173)** and **RA 10028** in its design and data handling
- The system shall be built using open-source technologies only (e.g., PHP or Python, MySQL or PostgreSQL)
- Uses open-source technologies
- The SMS notification feature depends on a third-party SMS gateway, making it subject to that provider's availability and limitations
- Data synchronization between hospital workstations and the central milk bank server requires a stable internet connection at both ends

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3. Functional Requirements

3.1 User Authentication

- The system shall authenticate staff using a username and password with role-based control (Administration, Nurse, Midwife), limiting access based on assigned roles.
- The system shall validate credentials and display an error message when incorrect credentials are input.
- The system shall send an activation code of the account via SMS verification.
- The system shall validate the activation of an account by requesting the verification code.
- The system shall allow users to log out securely.
- The system shall enable password reset and update functionality.
- The system shall allow account recovery via email/SMS verification.

3.2 Supsup Todo (Community/Mobile-Based Collection)

3.2.1 Donor Screening and Registration

- The system shall allow staff to record Supsup Todo milk collection sessions, capturing: Batch ID, donor reference (DTN), volume collected (mL), collection date, collecting staff, and collection method (community/barangay drop-off).
- The system shall enforce a maximum collection limit of 800 mL per donor per day and 30–240 mL per session, alerting staff if limits are exceeded.
- The system shall record bottling, labeling, and cold chain (cooler placement) information per collected batch.
- The system shall record the volume of pre-pasteurization samples taken (less than 5 mL) and log submission of samples to the City Hall laboratory.

3.2.2 Milk Collection

- The system shall allow staff to record milk donations for all three collection programs, capturing Batch ID, donor reference, volume (mL), collection date, and collecting staff.
- The system shall automatically flag batches that fail pre-pasteurization testing as disposed and prevent them from being dispensed.
- The system shall track collection methods (Hospital drop-off, Barangay milk drives, Home pickup service).

3.2.3 Pre-Pasteurization Lab Testing

- The system shall record pre-pasteurization laboratory test results (pass/fail) for each Supsup Todo batch, including the expected 2-week turnaround window.
- The system shall automatically flag batches with failed lab results as Disposed and prevent them from proceeding to pasteurization or dispensing.
- The system shall update batch status to Pending Pasteurization for batches that pass pre-pasteurization testing.

3.2.4 Pasteurization and Post-Pasteurization Testing

- The system shall allow staff to record pasteurization completion for Supsup Todo batches, logging the date and pasteurizing staff.
- The system shall record post-pasteurization sample submission and laboratory results (pass/fail).
- The system shall mark post-pasteurization passing batches as Ready for Dispensing and failing batches as Disposed.

3.3 Mom’s Act (Household/Home-Pickup-Based Collection)

3.3.1 Donor Screening and Registration

- The system shall allow staff to register Mom's Act donors, capturing personal information and pre-screening status (similar to Milky Way pre-natal screening records).
- The system shall record pre-screening results and consent records per donor.
- The system shall allow staff to search, view, and update Mom's Act donor profiles and mark them as active, inactive, or disqualified.

3.3.2 Pickup Request and Milk Collection

- The system shall allow staff to log a donor's pickup request, capturing the donor reference, requested pickup date and time, and assigned staff member.
- The system shall allow staff to record Mom's Act milk collection transactions after pickup, capturing: Batch ID, donor reference (DTN), volume collected (mL), pickup/collection date, collecting staff, and collection method (home pickup).
- The system shall enforce the 800 mL/day and 30–240 mL/session collection limits, alerting staff upon entry if exceeded.
- The system shall record sampling details (volume less than 5 mL) and log submission of samples to the City Hall laboratory.

3.3.3 Pre-Pasteurization Lab Testing

- The system shall record pre-pasteurization laboratory test results (pass/fail) for each Mom's Act batch.
- The system shall automatically flag batches with failed results as Disposed, preventing further processing or dispensing.
- The system shall update passing batches to Pending Pasteurization status.

3.3.4 Pasteurization and Post-Pasteurization Testing

- The system shall allow staff to log pasteurization completion for Mom's Act batches, recording the date and responsible staff.
- The system shall record post-pasteurization sample submission and lab results (pass/fail).
- The system shall mark passing batches as Ready for Dispensing and failing batches as Disposed.

3.4 Milky Way (Hospital-Based Collection)

3.4.1 Donor Pre-Screening and Registration

- The system shall allow hospital staff to register Milky Way donors using pre-natal screening records from partner hospitals, capturing personal information, pre-screening results, and consent status.
- The system shall allow hospital staff and milk bank staff to view, update, and search Milky Way donor profiles.
- The system shall allow staff to mark donors as active, inactive, or disqualified.

3.4.2 Hospital-Based Collection

- The system shall allow hospital staff to record Milky Way collection sessions at partner hospitals, capturing: Batch ID, donor reference (DTN), volume collected (mL), hospital name, collection date, collecting staff, and collection method (hospital-based).
- The system shall enforce the 800 mL/day and 30–240 mL/session limits with appropriate alerts.
- The system shall record bottling, labeling, and cold chain transfer details for batches transported to the milk bank.
- The system shall record sampling details (volume less than 5 mL) and log submission of samples to the City Hall laboratory.

3.4.3 Data Synchronization

- The system shall synchronize Milky Way collection records entered by hospital staff with the central milk bank database, ensuring records are accessible from both locations.
- The system shall support data sync between hospital workstations and the milk bank server.

3.4.4 Pre-Pasteurization Lab Testing

- The system shall record pre-pasteurization laboratory results (pass/fail) for each Milky Way batch.
- The system shall automatically flag failing batches as Disposed and prevent their dispensing.
- The system shall update passing batches to Pending Pasteurization status.

3.4.5 Pasteurization and Post-Pasteurization Testing

- The system shall allow staff to log pasteurization completion for Milky Way batches, recording the date and responsible staff.
- The system shall record post-pasteurization sampling and lab results (pass/fail).
- The system shall mark passing batches as Ready for Dispensing and failing batches as Disposed.

3.5 Inventory Management

- The system shall maintain a real-time inventory of all stored milk batches across all three programs.
- The system shall track each batch by status: Available, Reserved, Expired, or Discarded.
- The system shall monitor and display the volume (mL), expiration date, and storage conditions for each batch.
- The system shall generate automated alerts to staff when milk stock falls below a defined threshold or when batches are nearing expiration.

3.6 Beneficiary Management

- The system shall allow staff to register, search, view, and update beneficiary records, including the mother's personal details and infant's clinical information.
- The system shall link each dispensing record to the corresponding beneficiary profile.
- The system shall allow staff to log milk availability inquiries from beneficiaries who contact the hotline, capturing the beneficiary name, contact number, and date of inquiry.

3.7 Dispensing

- The system shall allow staff to record dispensing transactions, capturing: beneficiary information, volume dispensed (mL), dispensing fee per mL, bottle deposit fee, dispensing date, and releasing staff.
- The system shall verify that the required beneficiary documents are on file, Clinical Abstract, Pediatrician Prescription, and Cooler with Ice, before allowing a dispensing transaction to proceed.
- The system shall restrict dispensing to batches with a status of Ready for Dispensing only.
- The system shall prioritize beneficiaries based on medical urgency in the following order: Premature infants, NICU patients, and Immunocompromised babies.

3.8 SMS Notification

- The system shall automatically send an SMS notification to the registered beneficiary's mobile number when milk becomes available, logging the date and time of each notification sent.
- The system shall allow staff to log milk availability requests from beneficiaries who contact the hotline, and link those requests to the subsequent SMS dispatch.

3.9 Report Generation

- The system shall auto-generate Pasteurization Reports on daily, weekly, monthly, and annual bases, reflecting pre- and post-pasteurization outcomes per batch and program.
- The system shall auto-generate Dispensing Reports on daily, weekly, monthly, and annual bases, reflecting volume dispensed, fees collected, and beneficiary distribution.
- All generated reports shall support export and print functionality.

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4. Non-Functional Requirements

4.1 Performance Requirements

- The system shall respond within 3 seconds for standard operations.

- SMS notifications shall be dispatched within 30 seconds of a staff member triggering the notification in action

4.2 Security Requirements

- Passwords shall be encrypted.
- Role-based access control shall be enforced.
- The system shall maintain an audit trail of login attempts, including failed events with timestamps.
- All data transmissions between the browser and the server shall be conducted over a secure (HTTPS) connection.

4.3 Usability Requirements

- Interface shall be beginner-friendly.
- Interface shall be accessible via standard web browsers (Chrome, Firefox, Safari, Microsoft Edge, etc.) on desktop and mobile devices.
- Forms shall provide validation messages for missing or invalid fields, and the system shall display confirmation messages upon successful create, update, and delete operations.

4.4 Reliability Requirements

- The system shall maintain a minimum uptime of 99% during operating hours and perform automated daily database backups to prevent data loss.

4.5 Maintainability

- The system shall follow a modular architecture allowing independent module updates. Configuration parameters (e.g. dispensing fee, volume limits) shall be adjustable without modifying the source code.
- System administrators shall be able to update configuration parameters (e.g. dispensing fee/mL, session volume limits) without modifying source code.

4.6 Scalability

- The system shall support a growing number of registered users (e.g., donors, beneficiaries, and staff) without a decrease in performance.
- The system shall have a database structured efficiently to handle a large volume of records (e.g., milk batch, donor, dispensing) over time.
- The system shall allow the addition of new users/staff accounts without requiring architectural changes.

4.7 Regulatory Compliance

- The system must be compliant with Republic Act No. 10028 (Expanded Breastfeeding Promotion Act of 2009) and its implementing rules, including all rules related to the operation of human milk banks in the Philippines.
- In accordance with the Data Privacy Act of 2012 (RA 10173), the System will ensure that all personal information from donors, beneficiaries, and employees is collected legally and securely and is stored and processed securely in accordance with the individual's right to protect their personal information and to maintain their privacy.
- The system will provide complete and auditable records of all collected milk, laboratory testing, pasteurization processes, and dispensing activities performed according to the health regulatory standards.

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5. Use Case Diagrams

5.1 Use Case Diagram (UML)

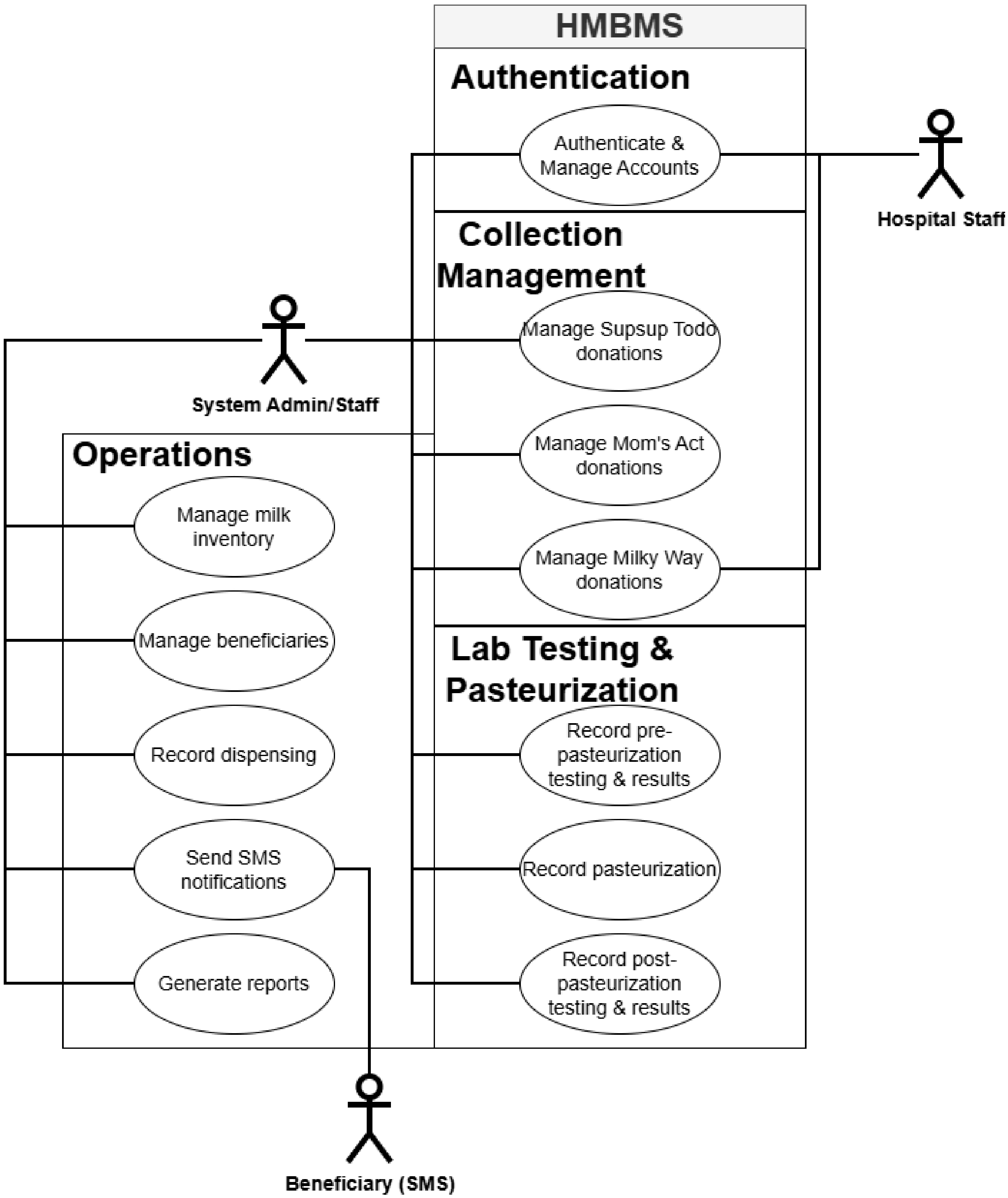

Figure 1. *HMBMS Use Case Diagram*

5.2 Use Case Description

5.2.1 Authentication

Use Case Name: Authenticate & Manage Accounts
Actor: System Admin/Staff, Hospital Staff
Precondition: The user is on the system login page.
Main Flow:
1. User enters their username and password.
2. System validates the credentials.
3. System checks the role-based access control (Admin, Nurse, Midwife).
4. System logs the user securely into their respective dashboard.

Postcondition: User is authenticated and granted appropriate access to the system.

5.2.2 Collection Management

Use Case Name: Manage Milky Way donations
Actor: Hospital Staff, System Admin/Staff
Precondition: Staff member is logged in, and the Milky Way donor is pre-screened or registered
Main Flow:
1. Staff selects to record a new Milky Way collection session.
2. Staff inputs the Batch ID, donor reference (DTN), volume collected (mL), hospital name, collection date, collecting staff, and method.
3. System checks and enforces the 800 mL/day and 30-240 mL/session volume limits.
4. Staff record the bottling, labeling, and cold chain transfer details.
5. Staff record the sampling details (volume <5 mL) and log the submission to the laboratory.
   Postcondition: The Milky Way milk collection record is saved, and the batch is ready for pre-pasteurization lab testing.

5.2.3 Operations

Use Case Name: Record dispensing
Actor: System Admin/Staff
Precondition: Staff is logged in, and the selected milk batch is in status "Ready for Dispensing".
Main Flow:
1. Staff initiates a dispensing transaction for a registered beneficiary.
2. System verifies that required documents (Clinical Abstract, Pediatrician Prescription, Cooler with Ice) are on file.
3. The system confirms beneficiary prioritization based on medical urgency (Premature, NICU, Immunocompromised).
4. Staff inputs the volume dispensed (mL), dispensing fee per mL, bottle deposit fee, dispensing date, and releasing staff.

5. The system saves the transaction and updates the real-time inventory.

Postcondition: The dispensing record is saved and linked to the beneficiary profile, and the inventory reflects the deducted milk volume.

5.2.4 Operations (Automated/Notification)

Use Case Name: Send SMS notifications
Actor: System Admin/Staff, Beneficiary (SMS)
Precondition: Pasteurized milk batches are updated to "Available" in the inventory, or a staff member logs a milk availability request.
Main Flow:

1. The system identifies that milk is available for dispensing.
2. The system automatically generates an SMS notification for the registered beneficiary.
3. The system dispatches the SMS to the beneficiary's mobile number.
4. The system logs the date and time the notification was sent.

Postcondition: The beneficiary receives an SMS notification, and the notification event is permanently logged in the system.

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6. System Design Diagrams

6.1 System Architecture Diagram

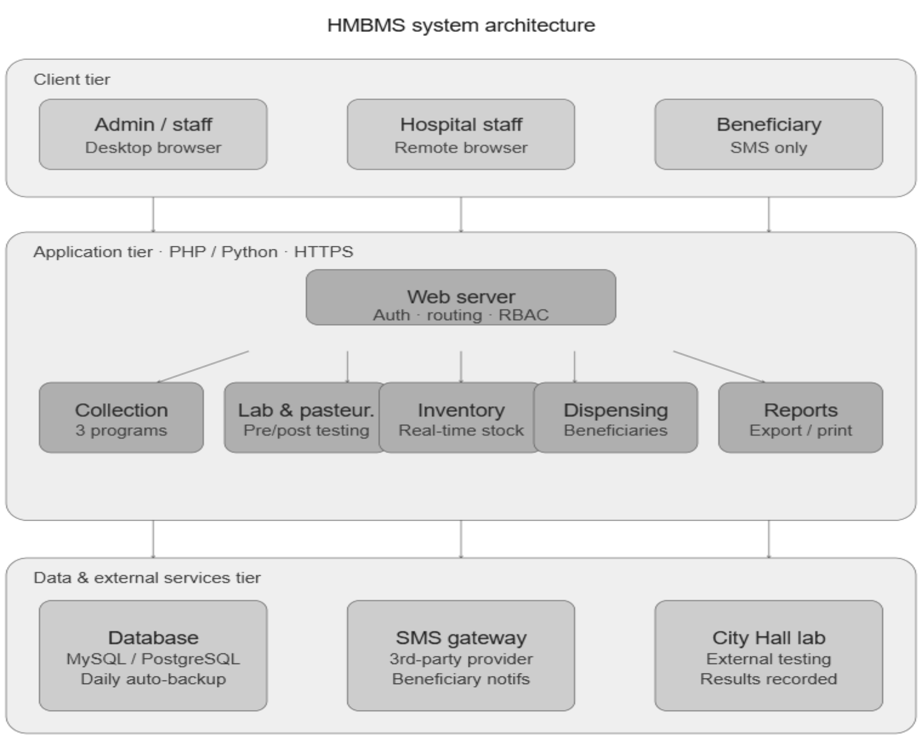

Figure 2. *HMBMS System Architecture Diagram*

A three-tier view of the system. The client tier has your three user types (admin/staff, hospital staff, beneficiary). The application tier shows the web server with its five major modules (collection, lab & pasteurization, inventory, dispensing, reports). The data tier shows the database, SMS gateway, and City Hall lab as external data dependencies.

6.2. Data Flow Diagrams

6.2.1 Data Flow Diagram (Level 0)

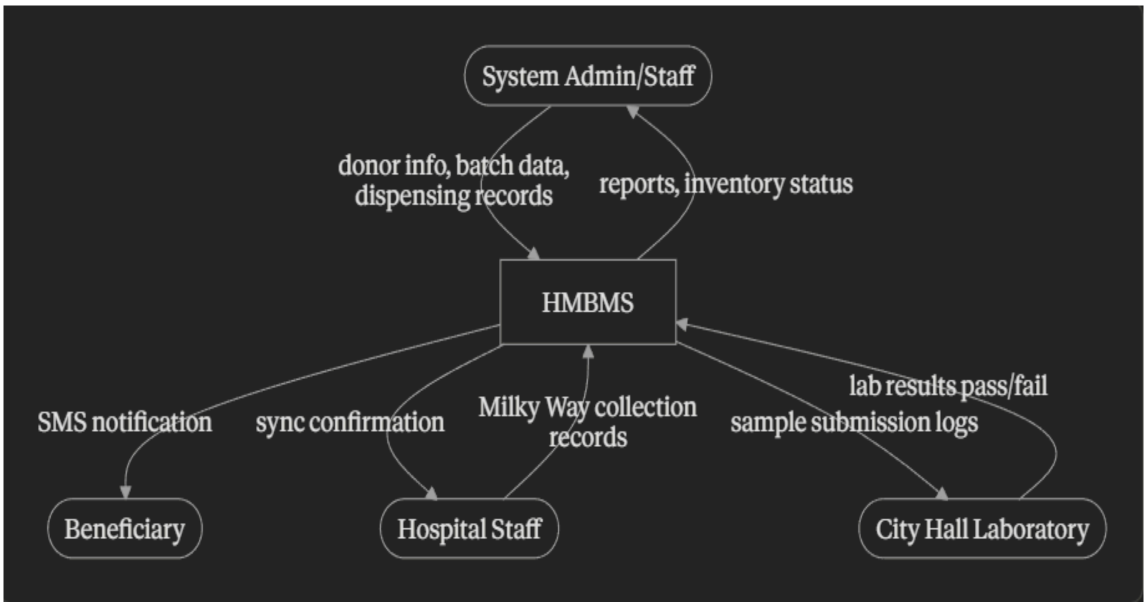

Figure 3. *HMBMS Data Flow Diagram (Level 0)*

The Level 0 DFD presents the Human Milk Bank Management System (HMBMS) as a single process, illustrating its boundaries and interactions with external entities. Four external entities interact with the system:

- **System Admin/Staff** — the primary users of the system, who input donor information, batch data, and dispensing records, and receive generated reports and inventory status updates in return.
- **Hospital Staff** — staff deployed at partner hospitals who submit Milky Way collection records into the system and receive synchronization confirmations.
- **Beneficiary** — recipient mothers and guardians who do not directly access the system but receive automated SMS notifications when pasteurized milk becomes available for dispensing.
- **City Hall Laboratory** — the external laboratory that receives sample submission logs from the system and returns pre- and post-pasteurization lab results (pass/fail) for each milk batch.

The context diagram establishes that all data entering and leaving the system passes through these four entities, and that the HMBMS serves as the central point of control for all milk bank operations.

6.2.2 Data Flow Diagram (Level 1)

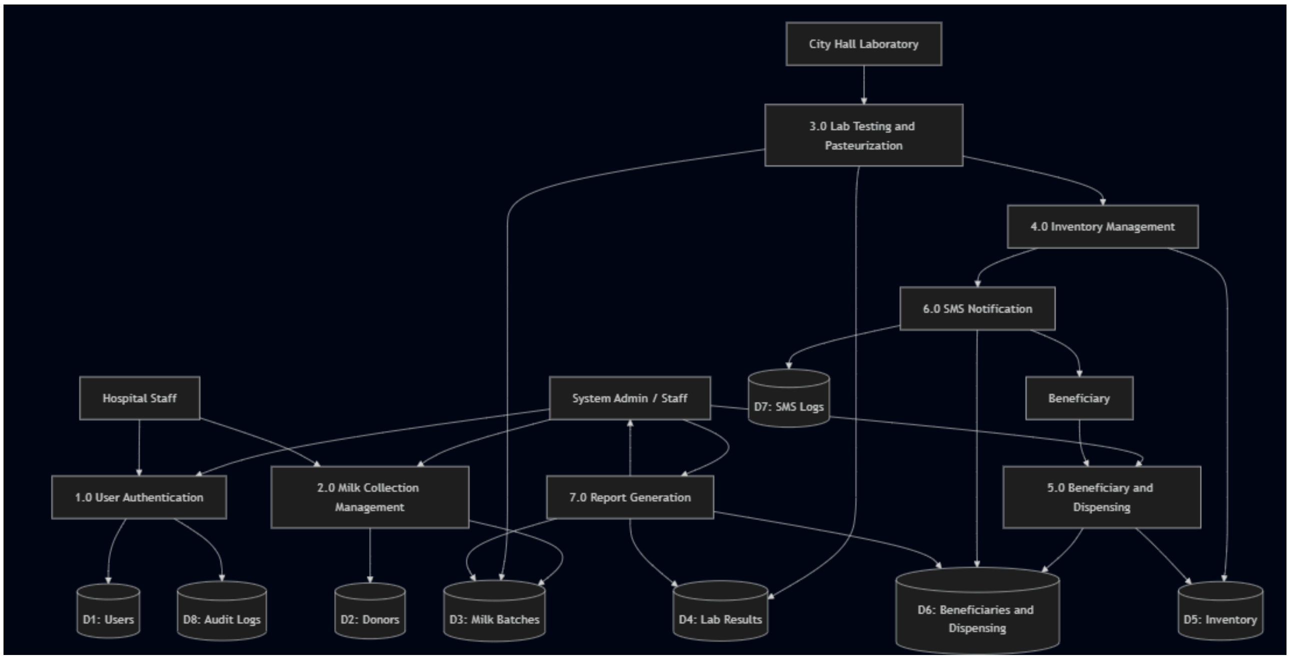

Figure 4. *HMBMS Data Flow Diagram (Level 1)*

The Level 1 DFD decomposes the HMBMS into seven processes, showing how data flows between external entities, internal processes, and data stores.

- **Process 1.0** — User Authentication System Admin/Staff and Hospital Staff submit their credentials to this process. Validated accounts are stored in the Users data store (D1), and all login activity is recorded in the Audit Logs data store (D8).
- **Process 2.0** — Milk Collection Management System Admin/Staff and Hospital Staff record milk donations from all three collection programs — Supsup Todo, Mom's Act, and Milky Way. Donor records are saved to the Donors data store (D2), and milk batch records are created in the Milk Batches data store (D3).
- **Process 3.0** — Lab Testing and Pasteurization The City Hall Laboratory submits test results to this process. Batch records are read from and updated in the Milk Batches data store (D3), and all test results are stored in the Lab Results data store (D4). Batches that pass all testing are forwarded to Process 4.0.
- **Process 4.0** — Inventory Management Cleared batches from Process 3.0 are tracked here. This process maintains the Inventory data store (D5) with each

batch's volume, status, and expiration details. When milk becomes available, it triggers Process 6.0.

- **Process 5.0** — Beneficiary and Dispensing System Admin/Staff record dispensing transactions and beneficiary information through this process. It reads and updates the Inventory data store (D5) to reflect dispensed volumes, and stores all transaction and beneficiary records in the Beneficiaries and Dispensing data store (D6).
- **Process 6.0** — SMS Notification Triggered by Process 4.0, this process sends an SMS alert directly to the Beneficiary and logs the notification in the SMS Logs data store (D7). It also references the Beneficiaries and Dispensing data store (D6) for contact information.
- **Process 7.0** — Report Generation Upon request by System Admin/Staff, this process reads from the Milk Batches (D3), Lab Results (D4), and Beneficiaries and Dispensing (D6) data stores to generate pasteurization and dispensing reports, which are returned to the requesting staff member.

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7. Database Design

7.1 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) for the Human Milk Bank Management System illustrates the logical relationships between the core data entities that support all milk bank operations. The diagram is structured around the central MILK_BATCHES entity, which connects directly or indirectly to all major modules of the system.

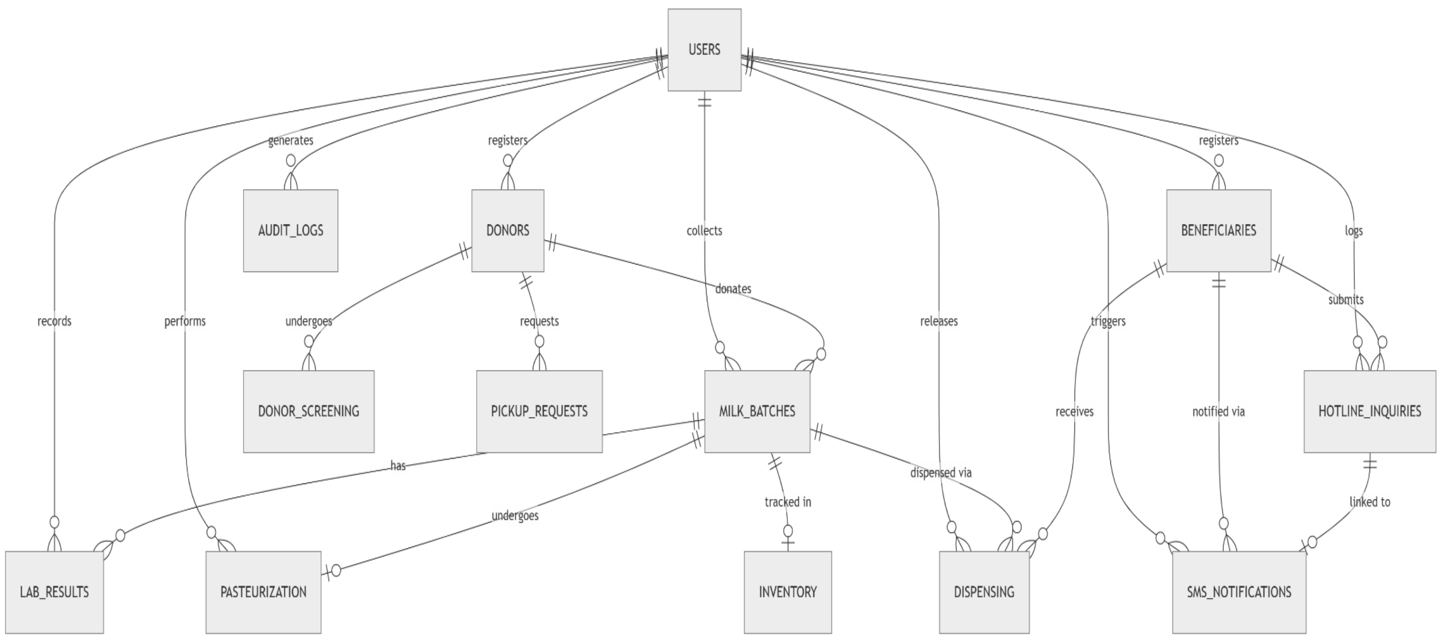

Figure 5. *HMBMS Entity Relationship Diagram*

- At the top level, the USERS entity represents all authenticated personnel (System Admin/Staff and Hospital Staff) and serves as the primary actor across multiple relationships. Users register DONORS and BENEFICIARIES, generate AUDIT_LOGS for accountability, and are linked to collection, pasteurization, and dispensing activities through foreign key references.
- The DONORS entity holds the profile and screening records of each lactating mother enrolled in the milk bank programs. Each donor undergoes DONOR_SCREENING prior to collection, and may submit PICKUP_REQUESTS (for Mom's Act donors). Donors are associated with one or more MILK_BATCHES through the collection process.
- The MILK_BATCHES entity is the operational hub of the system. Each batch is identified by a unique Batch ID and tracks the program it belongs to (Supsup Todo, Mom's Act, or Milky Way), the volume collected, collection date, and current status. Milk batches undergo LAB_RESULTS testing both before and after PASTEURIZATION. Passing batches flow into the INVENTORY and eventually into DISPENSING transactions.
- The BENEFICIARIES entity stores recipient infant and guardian information, including urgency level (premature, NICU, immunocompromised) and contact details. Beneficiaries receive milk through DISPENSING records and are notified via

SMS_NOTIFICATIONS. They may also submit inquiries through the HOTLINE_INQUIRIES entity, which links to both SMS_NOTIFICATIONS and beneficiary records.

- The AUDIT_LOGS entity records all system actions performed by users, including the affected record, action type, timestamp, and IP address, ensuring a full and auditable trail of all operations in compliance with data privacy requirements.

7.2 Database Schema

The database schema defines the physical structure of all tables in the HMBMS relational database. Each table corresponds to an entity identified in the ERD and includes defined field names, data types, and key constraints. The schema supports the full lifecycle of milk bank operations, from donor registration and milk collection through lab testing, pasteurization, dispensing, and notification.

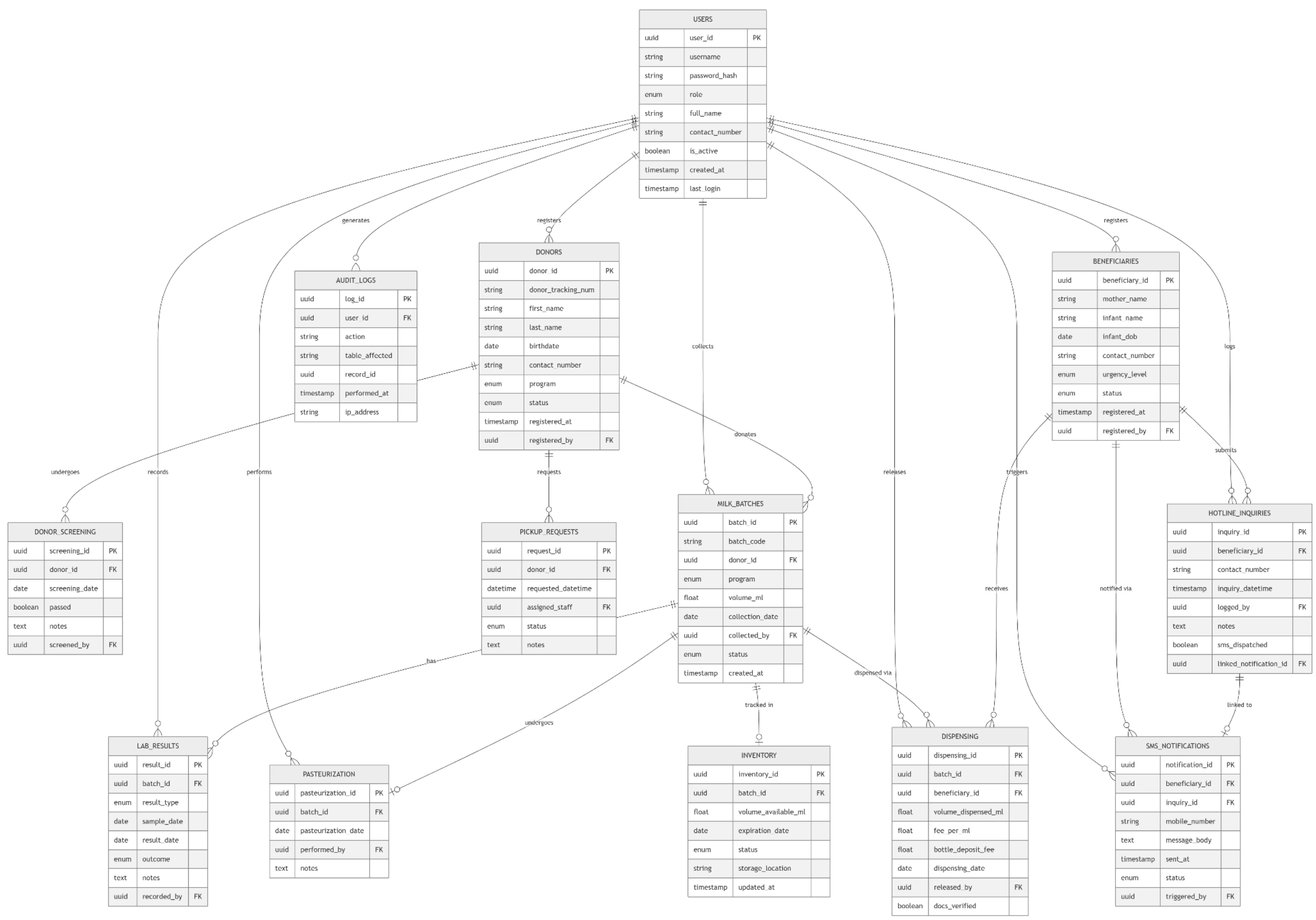

Figure 6. *HMBMS Database Schema*

- The USERS table serves as the authentication and role management foundation, storing encrypted passwords, role assignments (Admin, Nurse, Midwife), and login metadata. The DONORS table extends user-linked records with donor tracking numbers (DTN),

personal details, and program affiliation, while DONOR_SCREENING captures individual pre-screening results and consent status per donor.
- Milk collection data is stored in the MILK_BATCHES table, which records batch codes, volumes, programs, collection dates, and status fields (e.g., Pending Lab, Pending Pasteurization, Ready for Dispensing, Disposed). Each batch is linked to lab results through the LAB_RESULTS table, which stores result type (pre- or post-pasteurization), sample date, result date, pass/fail status, and the staff member who recorded the result. The PASTEURIZATION table captures pasteurization session details including date and responsible staff.
- The INVENTORY table tracks the real-time availability of each pasteurized batch, recording volume available, expiration date, storage location, and last-updated timestamp. The DISPENSING table records each dispensing transaction, linking the batch and beneficiary, and capturing volume dispensed, fees, and releasing staff.
- Beneficiary data is maintained in the BENEFICIARIES table, which includes infant name, birth date, contact number, and urgency level. Related entities include HOTLINE_INQUIRIES (logging milk availability calls) and SMS_NOTIFICATIONS (recording each automated SMS dispatch). The PICKUP_REQUESTS table supports Mom's Act workflow by storing scheduled pickup date, assigned staff, and request status.
- The AUDIT_LOGS table spans all operations, referencing user IDs and record IDs to maintain a complete, tamper-evident activity history for compliance with the Data Privacy Act of 2012 (RA 10173).

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8. Test Plan

8.1 Purpose

This Test Plan defines the testing strategy for the Human Milk Bank Management System (HMBMS). The objective of testing is to verify that the system meets all functional and non-functional requirements described in this SRS document, ensures the accuracy and integrity of all milk bank data, and operates reliably and securely in a clinical environment.

8.2 Test Objectives

- Verify that all system functionalities work as specified in the functional requirements
- Ensure accuracy and integrity of stored data across all modules
- Validate role-based access control for System Admin/Staff and Hospital Staff
- Confirm that collection volume limits are correctly enforced
- Verify that batch status transitions (lab testing, pasteurization, dispensing) are correctly handled
- Confirm acceptable system performance, usability, and SMS notification responsiveness
- Ensure compliance with data privacy and regulatory requirements

8.3 Test Scope

8.3.1 In Scope

- User authentication, account activation, and password management
- Milk collection recording for all three programs: Supsup Todo, Mom's Act, and Milky Way
- Pre-pasteurization and post-pasteurization lab result recording and batch status transitions
- Pasteurization completion logging
- Real-time inventory management and threshold alerts
- Beneficiary registration and management
- Dispensing transaction recording, document verification, and beneficiary prioritization
- Automated SMS notifications and hotline inquiry logging
- Automated report generation (daily, weekly, monthly, annual) and export/print
- Role-based access control enforcement
- Audit log integrity
- Data synchronization for Milky Way hospital collections

8.3.2 Out of Scope

- Hardware failure and physical infrastructure testing
- Network stress testing beyond defined concurrent-user thresholds
- Third-party SMS gateway internal testing
- Mobile application development (system is browser-based only)

8.4 Test Environment

Component	Description
Client	Web browsers: Google Chrome, Microsoft Edge, Brave
Server	Application server (local or cloud-hosted)
Database	MySQL or PostgreSQL
Operating System	Windows
SMS Gateway	Third-party SMS provider
Network	HTTPS-enabled internet connection required for data sync

8.5 Test Strategy

The system will be tested primarily using black-box testing, focusing on input/output validation without inspecting internal code. Testing will be conducted manually by the development team during system integration. Each test case will be executed in a controlled test environment that mirrors the production setup.

Where applicable, boundary value analysis will be applied, particularly for collection volume limits (30–240 mL per session, 800 mL per day) and time-based requirements (3-second response time, 30-second SMS dispatch). Negative test cases are included to verify that the system handles invalid inputs and unauthorized actions gracefully.

8.6 Test Cases

Test Case ID	Test Description	Precondition	Test Steps	Expected Result	Status
TC01	Login with valid credentials	User account exists in the system	1. Navigate to login page 2. Enter valid username and password 3. Click Login	User is authenticated and redirected to their role-based dashboard	Pass
TC02	Login with invalid credentials	User account exists in the system	1. Navigate to login page 2. Enter incorrect username or password 3. Click Login	System displays an error message indicating invalid credentials	Pass
TC03	Account activation via SMS verification	New account created by admin	1. Admin creates new account 2. The system sends activation code via SMS 3. User enters activation code	Account is activated and user can log in	Pass

TC04	Password reset via email/SMS	User account exists	1. Click 'Forgot Password' 2. Enter registered email/mobile 3. Receive and enter reset code 4. Set new password	Password is successfully updated	Pass
TC05	Secure logout	User is logged in	1. Click Logout button	Session is terminated; user is redirected to login page	Pass
TC06	Record Supsup Todo milk collection	Staff is logged in; donor is registered	1. Select Supsup Todo collection module 2. Input Batch ID, donor DTN, volume, date, method 3. Save record	Collection record is saved; batch is queued for pre-pasteurization lab testing	Pass
TC07	Enforce 800 mL/day collection limit	Staff is logged in; donor has existing collection	1. Attempt to record a collection that exceeds 800 mL total for the day 2. Save record	System displays an alert and prevents saving the record	Pass
TC08	Enforce 30–240 mL per-session limit	Staff is logged in	1. Attempt to enter a session volume outside 30–240 mL range 2. Save record	System displays a validation alert and blocks the submission	Pass
TC09	Record pre-pasteurization lab result (Pass)	Batch is in Pending Lab status	1. Open batch record 2. Enter pre-pasteurization lab results as Pass 3. Save	Batch status is updated to Pending Pasteurization	Pass

TC10	Record pre-pasteurization lab result (Fail)	Batch is in Pending Lab status	1. Open batch record 2. Enter pre-pasteurization lab results as Fail 3. Save	Batch is automatically flagged as Disposed; dispensing is blocked	Pass
TC11	Record pasteurization completion	Batch status is Pending Pasteurization	1. Open batch record 2. Log pasteurization date and responsible staff 3. Save	Pasteurization record is saved and batch is queued for post-pasteurization testing	Pass
TC12	Record post-pasteurization lab result (Pass)	Batch has been pasteurized	1. Enter post-pasteurization lab results as Pass 2. Save	Batch status is updated to Ready for Dispensing	Pass
TC13	Record post-pasteurization lab result (Fail)	Batch has been pasteurized	1. Enter post-pasteurization lab result as Fail 2. Save	Batch is automatically flagged as Disposed; dispensing is blocked	Pass
TC14	Register Mom's Act donor and record pickup request	Staff is logged in	1. Register donors with personal info and pre-screening 2. Log a pickup request with date and assigned staff 3. Save	Donor profile and pickup request are created and linked	Pass
TC15	Register Milky Way donor via hospital staff	Hospital staff is logged in	1. Input donor pre-natal screening records and consent 2. Save donor profile	Donor is registered and profile is accessible from both hospital and milk bank	Pass
TC16	Data synchronization – Milky Way collection	Hospital staff records a collection while connected	1. Hospital staff records Milky Way collection 2. Check record from milk bank workstation	Record appears on central milk bank database	Pass

TC17	Real-time inventory display	Batches with Ready for Dispensing status exist	1. Navigate to Inventory module 2. View all batches	All batches displayed with correct volume, expiration date, and status	Pass
TC18	Inventory alert – low stock or nearing expiration	Inventory thresholds are configured	1. Set a batch near threshold 2. Check alert status	System generates an automated alert to staff	Pass
TC19	Register beneficiary	Staff is logged in	1. Enter mother's personal info and infant's clinical data 2. Save record	Beneficiary profile is created and accessible for dispensing	Pass
TC20	Record dispensing transaction with required documents	Batch is Ready for Dispensing; beneficiary is registered	1. Initiate dispensing for registered beneficiary 2. Verify documents on file (Clinical Abstract, Rx, Cooler) 3. Input volume, fee, date, and releasing staff 4. Save	Dispensing record is saved; inventory is updated; record linked to beneficiary	Pass
TC21	Block dispensing when documents are incomplete	Batch is Ready for Dispensing	1. Initiate dispensing 2. Leave required documents unverified 3. Attempt to save	System blocks the transaction and displays document requirement alert	Pass
TC22	Beneficiary prioritization in dispensing	Multiple registered beneficiaries exist	1. View beneficiary queue 2. Check ordering	Premature infants appear first, followed by NICU patients, then immunocompromised babies	Pass

TC23	Automatic SMS notification on milk availability	Beneficiary is registered with mobile number; batch becomes available	1. Update batch status to Ready for Dispensing 2. Check SMS dispatch log	SMS is sent to beneficiary within 30 seconds; event is logged with timestamp	Pass
TC24	Log hotline milk availability inquiry	Staff is logged in	1. Open hotline inquiry form 2. Enter beneficiary name, contact number, and date 3. Save	Inquiry record is logged and linked to beneficiary profile	Pass
TC25	Generate daily pasteurization report	Pasteurization records exist for today	1. Navigate to Reports module 2. Select Pasteurization Report – Daily 3. Generate	Report displays all pre- and post-pasteurization outcomes for the current day	Pass
TC26	Generate monthly dispensing report	Dispensing records exist for the month	1. Navigate to Reports module 2. Select Dispensing Report – Monthly 3. Generate	Report displays volume dispensed, fees collected, and beneficiary distribution	Pass
TC27	Export and print report	A report has been generated	1. Click Export button 2. Click Print button	Report is exported successfully; print dialog opens	Pass
TC28	System response within 3 seconds	System is under normal operating load	1. Perform standard operations (login, form save, report generation) 2. Measure response time	All standard operations complete within 3 seconds	Pass
TC29	Role-based access control enforcement	Accounts with different roles exist (Admin, Nurse, Midwife)	1. Log in as each role 2. Attempt to access modules outside assigned permissions	Each role can only access permitted modules; unauthorized access is blocked	Pass

TC30	Audit log of login attempts	System is running	1. Perform successful and failed login attempts 2. Check audit log	Both successful and failed attempts are logged with timestamps and IP address	Pass

8.7 Entry and Exit Criteria

Entry Criteria

- All system modules are fully developed and deployed in the test environment
- Test environment is configured and operational (browser, server, database, SMS gateways sandbox)
- Test data (donor records, beneficiary profiles, milk batches) have been prepared
- Test cases have been reviewed and approved

Exit Criteria

- All critical cases (authentication, collection, dispensing, SMS notification) have passed
- All high-priority defects have been resolved and retested
- No outstanding blockers or critical failures remain
- Test results have been documented and reviewed by the project team

8.8 Requirements Traceability Matrix

The table below maps each functional and non-functional requirement to its corresponding use case, database table, and test case, ensuring full verifiability of the system.

Req ID	Requirement Description	Use Case	Database/Table	Test Case ID
FR-AUTH-01	User login with role-based access	UC01 – Authenticate & Manage Accounts	Users	TC01, TC02, TC05, TC29, TC30
FR-AUTH-02	Account activation via SMS	UC01 – Authenticate & Manage Accounts	Users	TC03
FR-AUTH-03	Password reset and update	UC01 – Authenticate & Manage Accounts	Users	TC04

FR-SUP-01	Supsup Todo milk collection recording	UC02 – Manage Supsup Todo Donations	MILK_BATCHES, DONORS	TC06
FR-SUP-02	Collection volume limit enforcement	UC02 – Manage Supsup Todo Donations	MILK_BATCHES	TC07, TC08
FR-LAB-01	Pre-pasteurization lab result (pass/fail)	UC03 – Record Pre-Pasteurization Testing	LAB_RESULTS, MILK_BATCHES	TC09, TC10
FR-PAS-01	Pasteurization completion recording	UC04 – Record Pasteurization	PASTEURIZATION, MILK_BATCHES	TC11
FR-LAB-02	Post-pasteurization lab result (pass/fail)	UC04 – Record Post-Pasteurization Testing	LAB_RESULTS, MILK_BATCHES	TC12, TC13
FR-MOM-01	Mom's Act donor registration and pickup request	UC05 – Manage Mom's Act Donations	DONORS, PICKUP_REQUESTS	TC14
FR-MIL-01	Milky Way donor registration (hospital)	UC06 – Manage Milky Way Donations	DONORS, DONOR_SCREENING	TC15
FR-MIL-02	Milky Way data synchronization	UC06 – Manage Milky Way Donations	MILK_BATCHES	TC16
FR-INV-01	Real-time inventory display	UC07 – Manage Milk Inventory	INVENTORY	TC17
FR-INV-02	Low stock and expiration alerts	UC07 – Manage Milk Inventory	INVENTORY	TC18
FR-BEN-01	Beneficiary registration and management	UC08 – Manage Beneficiaries	BENEFICIARIES	TC19
FR-DIS-01	Dispensing transaction with document check	UC09 – Record Dispensing	DISPENSING, INVENTORY	TC20, TC21
FR-DIS-02	Beneficiary prioritization in dispensing	UC09 – Record Dispensing	BENEFICIARIES, DISPENSING	TC22
FR-SMS-01	Automatic SMS notification on milk availability	UC10 – Send SMS Notifications	SMS_NOTIFICATIONS	TC23

FR-SMS-02	Hotline inquiry logging	UC10 – Send SMS Notifications	HOTLINE_INQUIRIES	TC24
FR-RPT-01	Automated pasteurization report generation	UC11 – Generate Reports	LAB_RESULTS, PASTEURIZATION	TC25, TC27
FR-RPT-02	Automated dispensing report generation	UC11 – Generate Reports	DISPENSING	TC26, TC27
NFR-PRF-01	System response within 3 seconds	All Use Cases	All Tables	TC28
NFR-SEC-01	Role-based access control	All Use Cases	Users	TC29
NFR-SEC-02	Audit trail of login events	All Use Cases	AUDIT_LOGS	TC30
NFR-SEC-03	HTTPS data transmission	All Use Cases	—	Security Review
NFR-REL-01	99% system uptime	—	—	Reliability Review